M. Chinni Krishna Reddy
192411192

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4303

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: **Total Marks: 30**

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
    <input type="text" name="name">
    <input type="email" name="email">
    <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
   <head>
      <title>Company</title>
   </head>
   <body>
      <div>
         <h1>Welcome
         <p>About us
      </div>
   </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
   <input type="text" name="username">
   <input type="password" name="password">
   <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I M. Chinni Krishna Reddy(192411192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

M. chinni Krishna reddy
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Estimated marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	21/25
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	22/25
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	18/20
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	14/15
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	9/10
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	5/5
Total						91/100

1. Online Symposium Portal

1. Identify appropriate HTML5 semantic elements missing

The missing semantic elements are <header>, <main>, and <section>. The form should also include proper labels for input fields.

2. Modified standards-compliant HTML5 code

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Symposium Registration</title>
</head>
<body>
   <header>
      <h1>Online Symposium Portal</h1>
   </header>

   <main>
      <section>
         <h2>Student Registration</h2>
         <form action="/register" method="POST">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name" required>

            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required>

            <button type="submit">Submit</button>
         </form>
      </section>
   </main>
</body>
</html>
```

3. Interpret the HTTP request generated

When the student submits the form, the browser sends an HTTP POST request to /register. The form data, including the name and email, is transmitted in the request body.

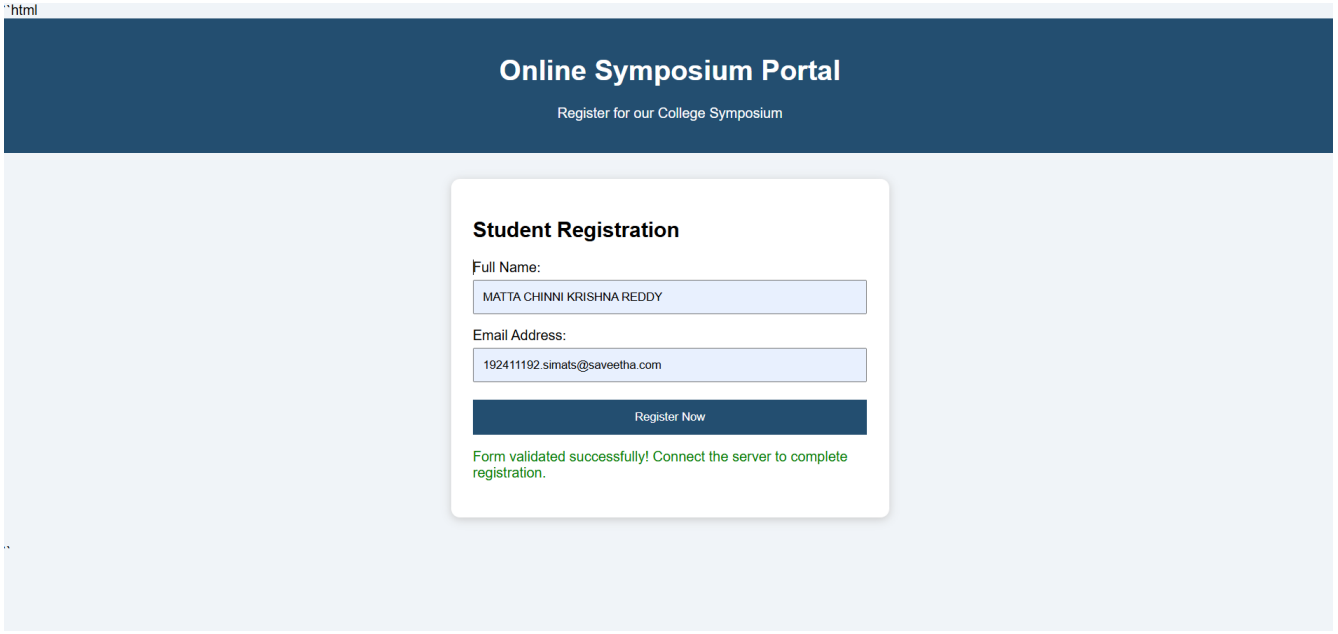
4. Analyze the HTTP response cycle

1. The browser sends the POST request to the server.
2. The server receives and validates the submitted data.
3. The server processes the registration.
4. The server returns an HTTP response with a status code, headers, and optionally a response body.
5. The browser displays the result to the student.

5. Improvements for usability and accessibility

- Use labels for all input fields.
- Mark required fields using required.
- Provide clear success and error messages.
- Use appropriate input types such as email.
- Ensure keyboard navigation and sufficient colour contrast.

```html
```


Online Symposium Portal
Register for our College Symposium
Student Registration
Full Name:
MATTA CHINNI KRISHNA REDDY
Email Address:
192411192.simats@saveetha.com
Register Now
Form validated successfully! Connect the server to complete registration.

```
```

2. Cross-Browser Compatibility Issue

1. Identify improper nesting and missing closing tags

The <h1> and <p> elements are missing their closing tags. The browser may attempt to repair the malformed HTML automatically.

2. Analyze how different browsers may interpret the structure

Browsers use HTML parsing rules to recover from errors. Although modern browsers generally repair this example similarly, malformed markup can cause inconsistent DOM structures and unexpected layouts.

3. Identify missing semantic elements

The code should use:

- <header> for the page heading.
- <main> for the main content.
- <section> for a meaningful content section

4. Corrected W3C-compliant HTML structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Company Website</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background: #f2f2f2;
    }

    header {
      background: #234e70;
      color: white;
      text-align: center;
      padding: 20px;
    }

    main {
      max-width: 800px;
      margin: 20px auto;
      padding: 20px;
      background: white;
    }

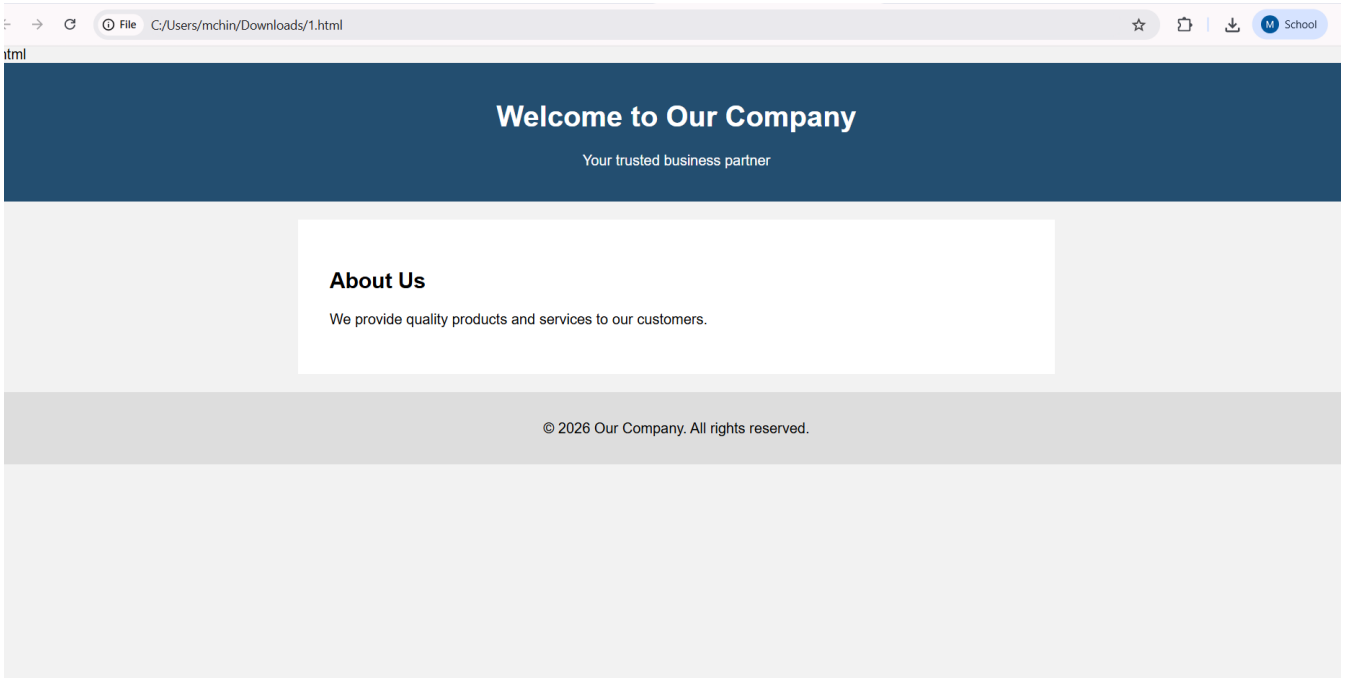
    section {
      padding: 15px;
    }

    footer {
      text-align: center;
      padding: 15px;
      background: #ddd;
    }
  </style>
</head>

<body>
  <header>
    <h1>Welcome to Our Company</h1>
    <p>Your trusted business partner</p>
  </header>
```

```
    <main>
        <section  aria-labelledby="about-heading">
            <h2 id="about-heading">About Us</h2>
            <p>We provide quality products and services to our customers.</p>
        </section>
    </main>

    <footer>
        <p>&copy; 2026 Our Company. All rights reserved.</p>
    </footer>
</body>
</html>
```



5. Techniques to ensure cross-browser compatibility

- Use valid HTML5 and CSS.
- Test the website in Chrome, Firefox, Edge, and Safari.
- Use responsive CSS and media queries.
- Avoid browser-specific features when possible.
- Validate the code using the official World Wide Web Consortium (W3C) validator.

3. Form Submission Failure

1. Identify issues in the form configuration

The main issue is type="button". This button does not submit the form automatically. The form also uses GET, which is inappropriate for transmitting passwords.

2. Analyze why the HTTP request is not triggered

The browser does not send a form submission request because the button is not a submit button. Without JavaScript that handles the click, the form remains unsubmitted.

3. Interpret the role of GET method

The GET method sends form data as URL query parameters. It is suitable for retrieving information, but it should not be used to transmit passwords because sensitive data can appear in URLs, browser history, and logs.

4. Corrected implementation

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Login Form</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #eef2f7;
      margin: 0;
    }

    main {
      max-width: 350px;
      margin: 60px auto;
      padding: 25px;
      background: white;
      border-radius: 10px;
      box-shadow: 0 2px 10px #ccc;
    }

    label {
      display: block;
      margin-top: 15px;
    }

    input {
      width: 100%;
      padding: 10px;
      margin-top: 5px;
      box-sizing: border-box;
    }

    button {
      width: 100%;
```

```
            padding: 12px;
            margin-top: 20px;
            background: #234e70;
            color: white;
            border: none;
            cursor: pointer;
        }

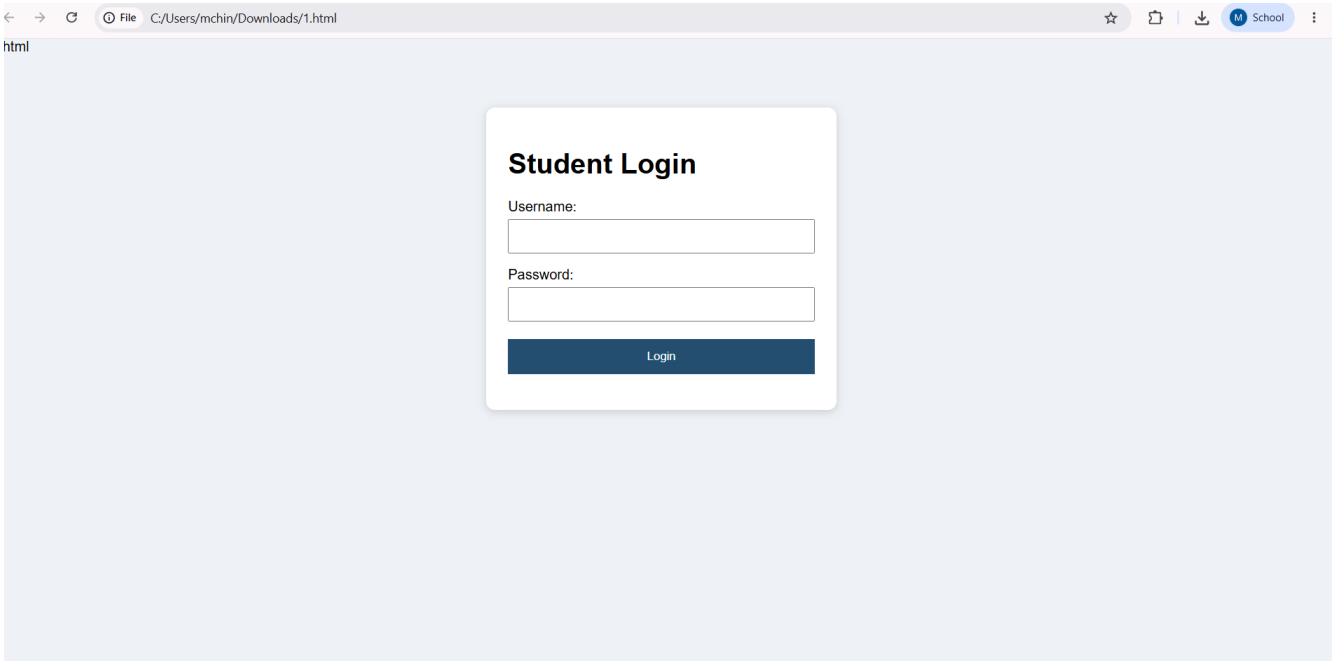
        button:hover {
            background: #16324f;
        }
    </style>
</head>

<body>
    <main>
        <section>
            <h1>Student Login</h1>

            <form action="/submit" method="POST">
                <label for="username">Username:</label>
                <input type="text" id="username" name="username"
                    autocomplete="username" required>

                <label for="password">Password:</label>
                <input type="password" id="password" name="password"
                    autocomplete="current-password" required>

                <button type="submit">Login</button>
            </form>
        </section>
    </main>
</body>
</html>
```



Result: The login form validates required fields and submits the username and password through an HTTP POST request to /submit.

Important: The form requires a working server endpoint at /submit to process the login. Use HTTPS to protect credentials during transmission. POST alone does not encrypt the data.